Carry / Levered LP Vault — Contract Overview

Grant and audit reference for `LeveredLpVault` on **Robinhood Chain Testnet (chain id 46630)**.

Product vault (PROXY): `0x24B799026AA7baa6856bCFD00c1Be1f583A83e35` (UUPS)
Implementation: `0x7BeED7F65A9D4aF4E67553d984800F6268a3778E` (`version()` = **`v3-lender-pool`**)
Site: usecarry.io
Superseded v2.5 PROXY (history only): `0xc80108649B3ba2e5B040c79DDE3af0cB979b72bd`
v1 grant/history reference (not in use): `0x72A053120f03B10c5506d2325f92F59B0FfC36c8`

Full pack: `../../CARRY_CONTRACT.md`. Accounting: `../../docs/ACCOUNTING_V3.md`.

This overview describes the source on `main`. The live implementation predates three merged changes: the EIP-170 fit (#3), rolling positions (#16) and `StrategyBoostHook` (#23). They reach testnet with the next owner-signed upgrade.

Lender rate note. The Morpho rate here is an owner-set testnet stub, not a promised APY. The mainnet lender return is being reworked to Morpho yield (floating) + Carry's share of fees, with no fixed rate or floor (issue #20, PR #21).

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1. Overview

Carry pairs a stock-token holder (**junior**) with a USDG lender (**senior**) into a **50/50 book**: junior posts MSTR, lenders supply matching USDG from a shared pool. Equity tracks the stock while the book accrues LP fees. This vault is the **custody and accounting layer**.

- Junior deposits MSTR → **Idle** until pool USDG matches, then **Active**. Product states: Idle / Active / Boosted / Closed.
- Term is **7 days**; at term a position rolls rather than closing, and only the junior settles a healthy one. A position more than 32% below entry stops rolling; 48h after the last term boundary anyone can settle it.
- Senior accrual = principal × change in a floating rate index driven by `morphoRateWad` (testnet stub, cap **≤5%**).
- Fee waterfall runs only at `claimFees` / `earlyExit` / `settle` (no cut on fee-in).
- **Early exit** (first term only) cover: Wallet / IdleCarry / SellShares (junior only, NO backstop). **Settle:** backstop first, then junior cover; **default = IdleCarry**.
- Lenders hold shares in **one USDG pool**. Withdraw is instant when idle USDG covers it, else FIFO queue + `claimWithdraw()`.

Contract snapshot

Item	Value
Contract	`LeveredLpVault` / `LeveredLpVaultV2` (UUPS PROXY product CA)
Tokens / oracle	Immutable `mstr`, `usdg`; immutable `IPriceOracle`
Term / Morpho rate	Immutable `term` ≤ 7 days; `morphoRateWad` ≤ 5% (testnet stub) drives a floating `rateIndex`
Fee cuts	Treasury base 20% (Boosted cut from `boostHook`, capped at base) + senior perf 20%
Accounting	One lender pool (`totalSeniorAssets`), non-transferable internal shares; yield compounds into the pool
Caps	Min $25; senior pool uncapped; junior market $25k; **no per-wallet cap** on any market
Unpaid accrual	Carried on position when fees < accrual
Pause / reentrancy	Owner pause gates deposits; `nonReentrant` on value paths
External LP	No PoolManager approvals; `joinPool` reverts
Runtime size	`main` ~23.4KB (under EIP-170); live impl ~29.6KB (testnet only)

Inventory is held inside the vault. `mark()` reports 2× equity vs an unlevered 50/50 √p comparison without moving inventory.

2. Actors

Actor	Role	Key actions
Junior (stock depositor)	Posts MSTR	`depositJunior`, `withdrawUnmatched`, `earlyExit`, `claimFees`, `settle`, gasCredit, auto-compound
Senior (lender)	Posts USDG into the pool	`depositSenior`, `withdrawSenior` (instant or queued), `claimWithdraw`
Owner / protocol	Pause + upgrade + knobs	`pause`, `unpause`, UUPS upgrade, caps, `setBoostHook`, `setMorphoRate`; cannot pull MSTR/USDG principal
Boost hook	Maps STRATEGY stake to a treasury cut	`StrategyBoostHook.treasuryCutOf` (Level I-V, placeholder 17.5/16.25/13.75/10/5%)
Fee source	Accrues LP fees	`accrueLpFee` / `MockFeePool.payFee`
Anyone	Keepers	`settle` a position that stopped rolling, after the 48h grace

3. Lifecycle

1. **Deposit stock** — `depositJunior(mstrAmount)`. Partial match OK; residual stays Idle.
2. **Lend USDG** — `depositSenior(amount)`. Mints pool shares; fills the withdraw queue, then FIFO-matches Idle juniors. First match snapshots the rate index, locks the treasury cut and starts the term clock.
3. **Active** — Accrual follows the floating rate index. At term the position rolls.
4. **Fee accrual** — `accrueLpFee` pulls raw USDG onto the position (no cut on fee-in).
5. **claimFees / early exit / settle** — Maker waterfall only here. Early exit: junior cover only (NO backstop). Settle: backstop first, then junior cover; default IdleCarry.

6. **Lender exit** — `withdrawSenior(assets)` instant when idle covers and the queue is empty; else queued, and `claimWithdraw()` pays the locked shares' value once filled.

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4. Function reference

Admin

Function	Who	Preconditions	Effects / funds
`pause()` / `unpause()`	owner	—	Gates new deposits
`rescueToken(token,to,amount)`	owner	`token ∉ {mstr,usdg}`	Transfers non-principal token
Caps / boost / rate	owner	—	`setDepositCaps`, `setMinDepositUsdg`, `setBoostHook`, `setMorphoRate`, `setTreasuryCuts`, …

Junior

Function	Who	Preconditions	Effects / funds
`depositJunior(mstrAmount)`	anyone (when unpaused)	min notional	Pulls MSTR; enqueues Idle; tries match
`withdrawUnmatched(positionId)`	position owner	Idle	Returns all MSTR; gasCredit refund
`earlyExit(positionId[, mode])`	position owner	Matched, first term	Maker waterfall; cover Wallet / Idle / SellShares

Senior

Function	Who	Preconditions	Effects / funds
`depositSenior(amount)`	anyone (when unpaused)	min	Mints pool shares; fills queue; FIFO match
`withdrawSenior(assets)`	lender	`≤` free share value	Instant if idle covers and queue empty; else queued
`claimWithdraw()`	lender	queued request filled	Pays filled USDG

Fees / settle / views

Function	Who	Preconditions	Effects / funds
`accrueLpFee` / `claimFees` / `fundBackstop`	see full pack	—	Raw fee-in; waterfall on claim
`settle(positionId[, mode])`	junior; anyone once a position stopped rolling	term elapsed	Default IdleCarry
`freePrincipal` / `convertToAssets` / `claimableWithdraw` / `currentRateIndex` / `isMatched` / `mark`	view	—	Pool, queue, rate, MTM
`joinPool(bytes)`	anyone	—	Reverts `ExternalLpForbidden`

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5. Accounting

```
accrual = unpaidSeniorAccrual
        + seniorPrincipal * (rateIndexNow - rateIndexSnapshot) / WAD
```

Waterfall at claim/exit/settle:

```
seniorFloor = min(gross, accrual)
treasury    = min(left, gross × cut)   // base 20%; Boosted cut from boostHook, capped at base
seniorPerf  = min(left, gross × 0.20)
junior      = remainder
```

Senior USDG goes to the lender pool (share price rises). See `ACCOUNTING_V3.md`. The `seniorFloor` / rate-index part is what the planned no-fixed-rate rework removes.

6. Security model

Control	Behavior
Pause	Blocks deposits only; exits/settle/withdraw remain
Owner cannot	Pull MSTR/USDG principal via rescue
Reentrancy	`nonReentrant` on value paths
Fee-on-transfer	Rejected on pulls
External LP	Forbidden
Boost hook	Result capped at base cut; a bad hook can only lower Carry's take
Impl size	`main` under EIP-170; live testnet impl over it until upgraded

Details: `SECURITY_V3.md`.

7. Carry UI → contract mapping

Carry screen / control	Vault call	Notes
Deposit MSTR	`depositJunior`	Idle if no USDG; min $25
Badge **Idle** / **Active** / **Boosted** / **Closed**	match + settled + boost flags	Not Open/Settled
Lend USDG	`depositSenior`	Pool shares; fills queue then FIFO match
Withdraw Early	`earlyExit(id, mode)`	Wallet / Idle / SellShares
Settle	`settle(id[, mode])`	Default IdleCarry
Lender withdraw	`withdrawSenior(assets)` / `claimWithdraw()`	Instant or queued

8. Testnet addresses (Robinhood Chain Testnet 46630)

Item	Value
Network	Robinhood Chain Testnet
Chain ID	**46630**
RPC	`https://rpc.testnet.chain.robinhood.com`
Explorer	`https://explorer.testnet.chain.robinhood.com`
Site	usecarry.io

Product vault (UUPS proxy)

Name	Address
LeveredLpVault (PROXY)	`0x24B799026AA7baa6856bCFD00c1Be1f583A83e35`
implementation (`v3-lender-pool`)	`0x7BeED7F65A9D4aF4E67553d984800F6268a3778E`
CarryMath	`0x86b2f56e58c38f19559f28e1463f2dcac1c59344`
CarryPool	`0xf1d5cb8c25bc1b2ed147cb5f2f63ebfacddbdc07`
CarryCover	`0x99f1702044603df846a038d4c1fb8563a57e1ebd`
v2.5 PROXY (superseded)	`0xc80108649B3ba2e5B040c79DDE3af0cB979b72bd`
v1 grant reference (not in use)	`0x72A053120f03B10c5506d2325f92F59B0FfC36c8`

Supporting contracts

Name	Address
mockMstr	`0x762019309B536bbb89577422FaaFBeC9659f8728`
mockUsdg	`0x25030Bff74764aD72b912276a603717DB1C00644`
mockOracle ($100 WAD)	`0xc74Af7E23A2B4b46B5Fe05E5c7c5ec0BB5dbc5B7`
mockFeePool	`0xa4122524aD97Aab05cAC7F99c04Cd060D02F0771`
vaultOwner	`0xcA44F2dbB2D43b93b39F01B88E0f0e2966983c57`

Live deploy settings: term **7 days**, `morphoRateWad` testnet stub (≤5%), treasury base **20%**, oracle **$100** WAD, min **$25**, senior pool uncapped, junior market **$25k**, no per-wallet cap.

Source paths

Path	Role
`contracts/src/LeveredLpVault.sol`	Vault
`contracts/src/StrategyBoostHook.sol`	Boosted treasury cut from STRATEGY stake
`CARRY_CONTRACT.md`	Full contract pack
`CARRY_TESTNET.md`	Deploy addresses and txs
`docs/ACCOUNTING_V3.md`	Live accounting model
`docs/testnet-runs/`	Dated on-chain test passes